Evaluating the Predictive Power of Audio Features on Spotify Popularity

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1. Introduction

The music industry is highly competitive, and understanding what makes a song popular is somewhat complicated. This project focuses on predicting a track's popularity score (ranging from 0 to 100) using measurable audio features such as energy, tempo, danceability, and acousticness.

The motivation behind this project is to determine whether the intrinsic sound characteristics of a song are sufficient to predict its success. This task presents a challenge because popularity is influenced by many external factors that are not captured in audio data, like artist reputation, marketing strategies, and social media exposure.

To address this problem, a machine learning pipeline was developed using a large dataset of Spotify tracks. A baseline Linear Regression model was implemented to evaluate simple relationships, followed by a Random Forest Regressor to capture more complex, non-linear patterns.

2. Data

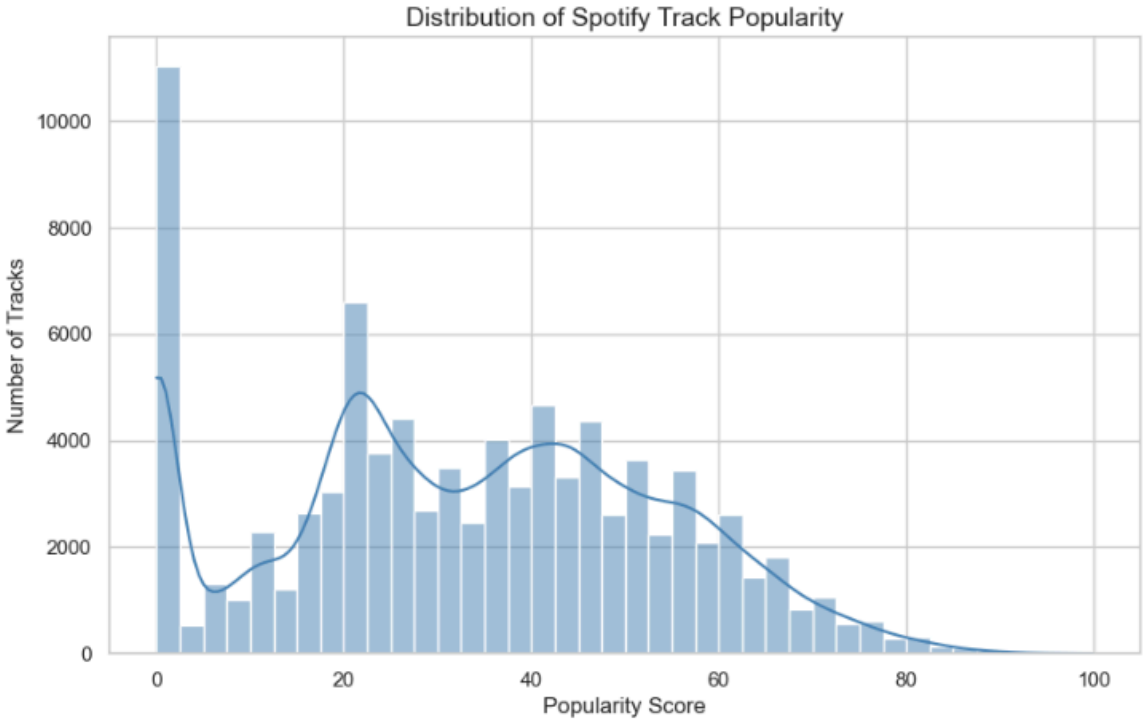
The dataset used in this project is the Spotify Tracks Dataset, which contains metadata and audio features for songs. The original dataset contained approximately 114,000 entries. After removing duplicate track IDs and unnecessary columns, the dataset was reduced to approximately **89,741** unique tracks.

Each track includes numerical features:

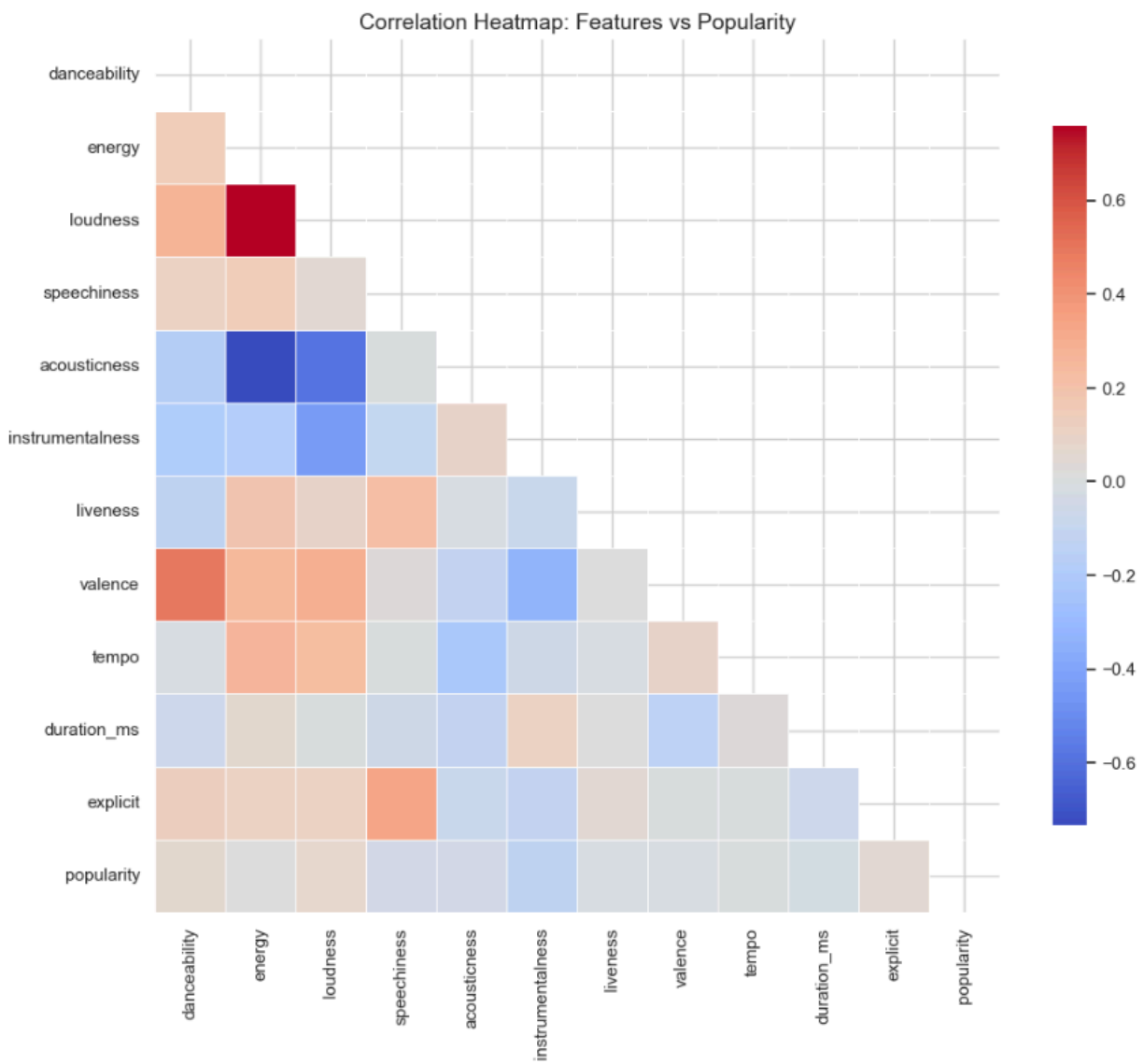
- danceability
- energy
- loudness
- speechiness
- acousticness
- instrumentalness
- liveness
- valence
- tempo
- duration (ms)
- explicit indicator

The target variable is **popularity**, which is measured on a scale from 0 to 100.

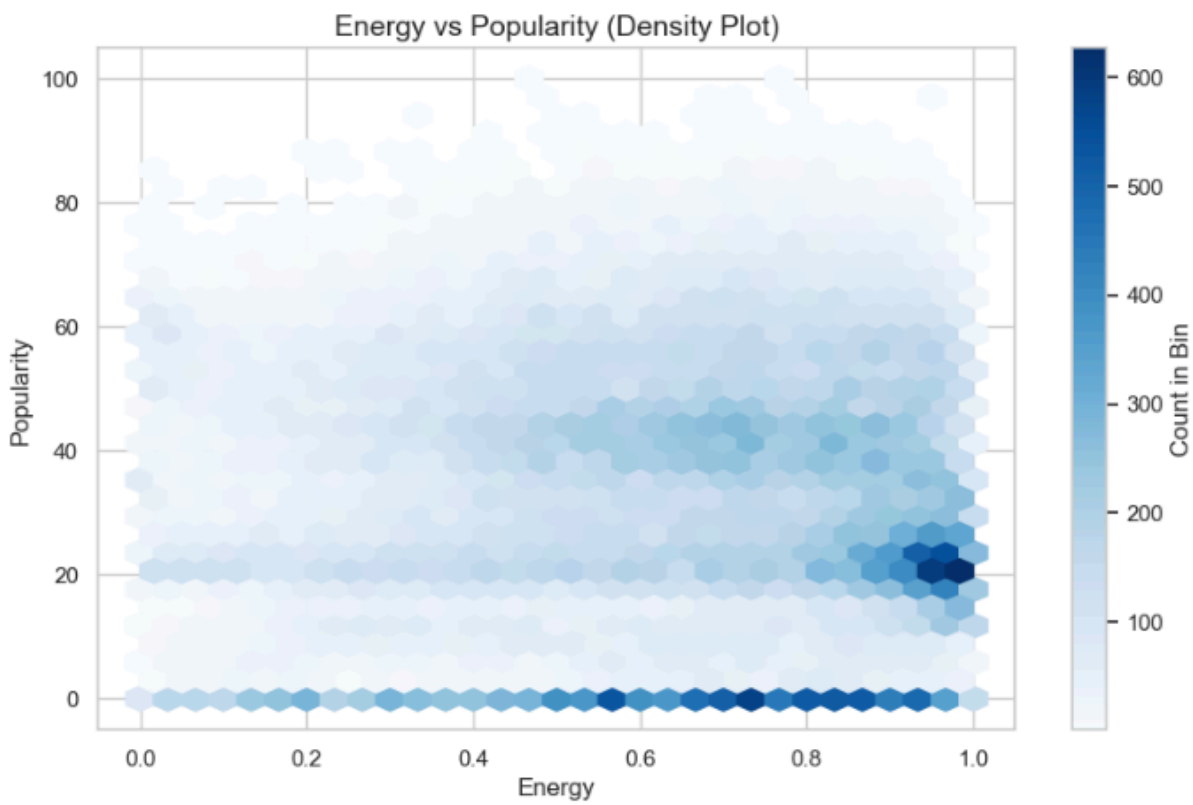
2.1 Exploratory Data Analysis



The distribution of track popularity is heavily skewed toward lower values, with a pronounced concentration near zero. This shows that a large proportion of songs receive little to no engagement, while highly popular tracks are relatively rare. The long right tail suggests that only a small subset of songs achieve high popularity scores, highlighting the imbalanced nature of success in the music industry. This skewed distribution also makes prediction more challenging, as the model must account for a large number of low-popularity observations.



The correlation heatmap illustrates relationships between audio features and popularity. While certain features, like energy and loudness, exhibit strong positive correlation with each other (indicating overlapping acoustic characteristics), none of the features demonstrate a strong linear relationship with popularity. Most correlations with popularity are weak or near zero, suggesting that no single audio feature is a strong standalone predictor. This supports the hypothesis that popularity is influenced by a combination of factors, many of which are not captured within the dataset.



The density plot of energy versus popularity shows that tracks are widely distributed across all popularity levels regardless of energy values. High-density regions are concentrated in mid-energy and low-to-moderate popularity ranges, indicating that even energetically “strong” songs are not guaranteed to be popular. The absence of a clear upward or downward trend confirms that energy alone does not have a strong predictive relationship with popularity, reinforcing the need for more complex models and other features.

Overall, the exploratory analysis suggests that popularity is weakly related to individual audio features, motivating the use of non-linear models like Random Forest.

3. Preprocessing

Several preprocessing steps were applied to prepare the dataset for modeling:

- **Duplicate Removal:** Duplicate entries based on `track_id` were removed to ensure that each observation represents a unique track and to prevent bias from repeated data.
- **Feature Selection:** Only numerical audio features were selected for modeling. Categorical variables like artist names and track titles were excluded to keep the model simple and focused on numerical inputs.
- **Missing Values:** The dataset contained no missing values; therefore, no imputation or data-filling techniques were required.
- **Feature Scaling:** Feature scaling was not applied. Tree-based models, like Random Forest, are invariant to feature scale, and the baseline Linear Regression model performed within expected ranges without scaling.

The dataset was then divided into training and testing subsets using an 80/20 split. A fixed random seed (`random_state = 42`) was used to ensure reproducibility of results.

4. Methods

4.1 Linear Regression

Linear Regression was used as a baseline model to evaluate whether a simple linear relationship exists between the selected audio features and track popularity. This model assumes that the target variable can be expressed as a weighted sum of the input features.

As a baseline, Linear Regression provides a simple and interpretable benchmark, allowing for comparison against more complex models. Its performance helps determine whether linear assumptions are sufficient for this prediction task.

4.2 Random Forest Regressor

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions to improve accuracy and reduce overfitting. Unlike Linear Regression, it is capable of capturing non-linear relationships and complex interactions between features.

This is especially crucial in this project, as exploratory analysis indicated weak linear relationships between individual audio features and popularity.

The model was configured with:

- **100 trees** (`n_estimators = 100`)
- **Parallel processing enabled** (`n_jobs = -1`)

Random Forest was selected due to its robustness, ability to handle high-dimensional tabular data, and effectiveness in modeling complex patterns without requiring extensive feature engineering.

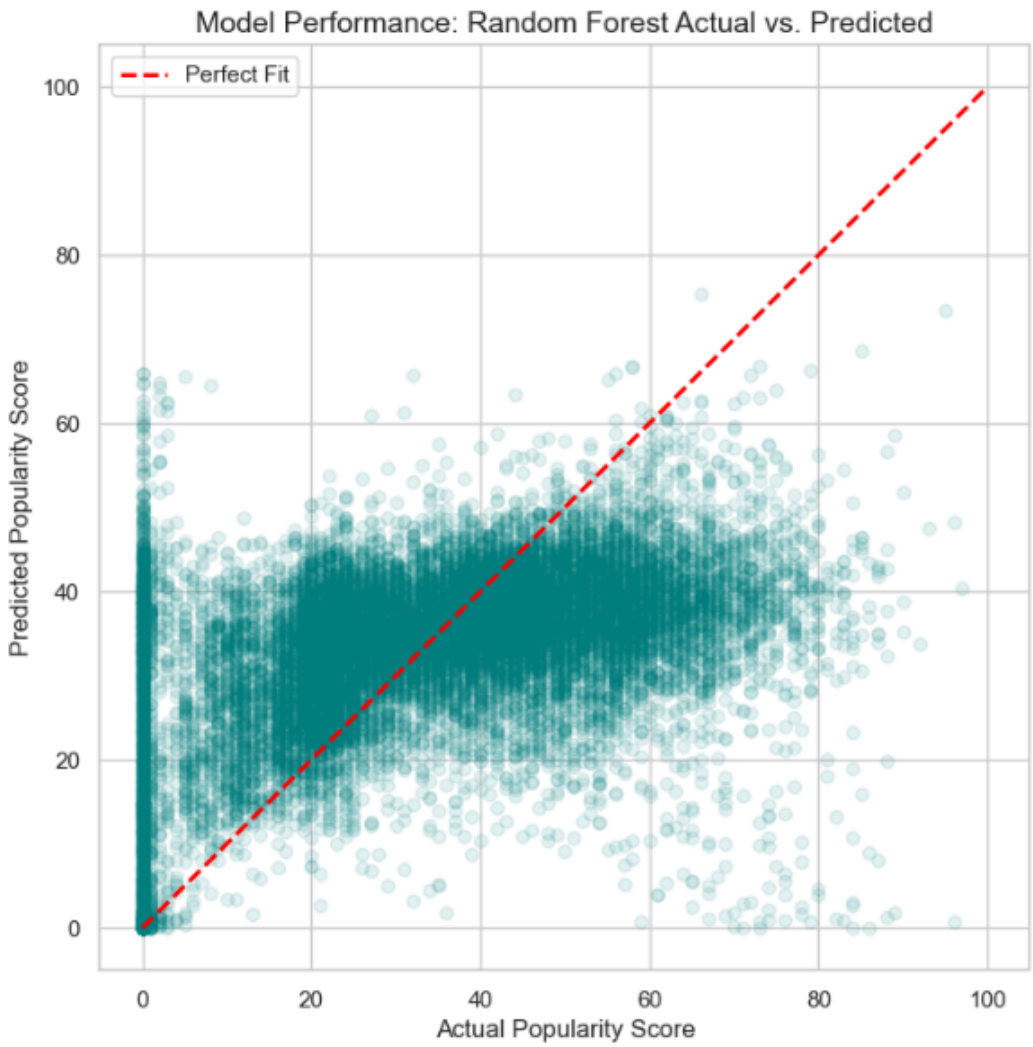
5. Results

5.1 Model Performance

Model	RMSE	R^2 Score
Linear Regression	20.13	0.0300
Random Forest	17.76	0.2453

The Random Forest model significantly outperforms Linear Regression, achieving a lower RMSE and a substantially higher R^2 score. This shows that the relationship between audio features and track popularity is non-linear, and that more flexible models are better suited for capturing underlying patterns in the data.

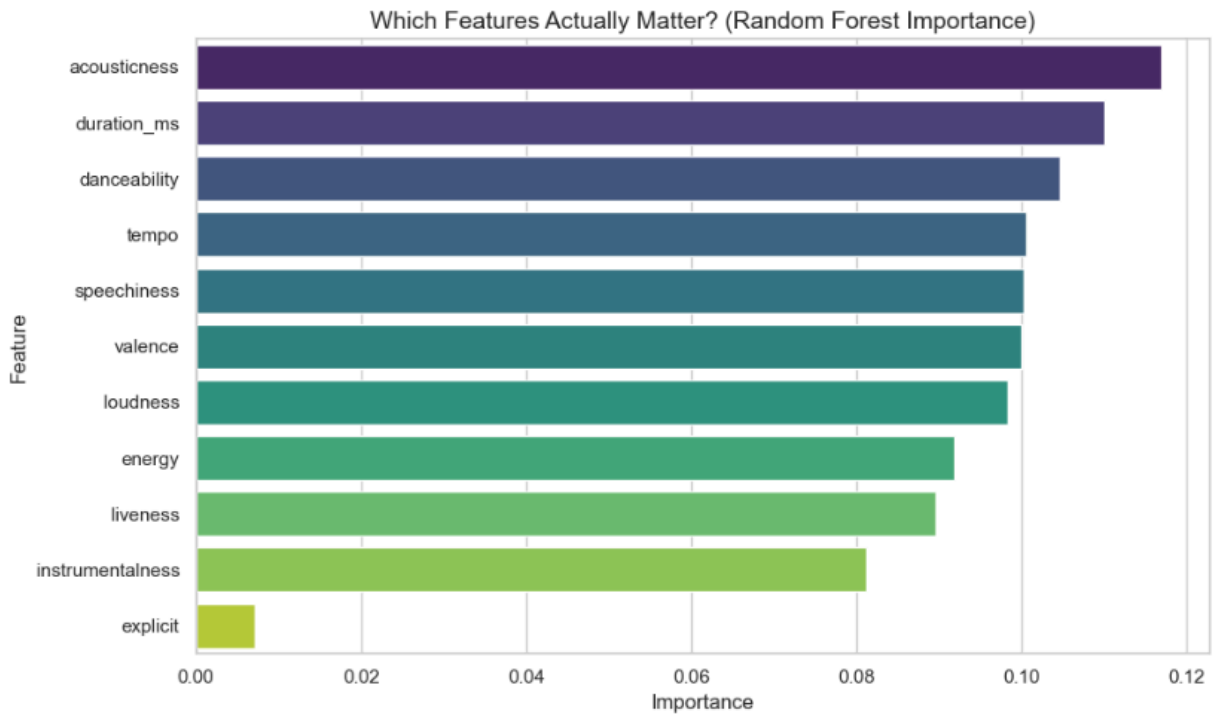
Despite this improvement, the overall R^2 value remains relatively low (~0.245), meaning that only about 24.5% of the variance in popularity is explained by the model. This shows that while audio features do play a role, they are not the primary drivers of popularity.



The scatter plot compares the model's predicted popularity scores against the actual values. Ideally, predictions would align along the diagonal "perfect fit" line. The points are widely dispersed, indicating prediction error.

A noticeable pattern is that predictions tend to cluster around mid-range values, with the model underestimating highly popular tracks and overestimating low-popularity tracks. This shows regression toward the mean, where the model struggles to capture extreme values.

This behavior is consistent with the relatively low R^2 score and further supports the conclusion that audio features alone are insufficient to accurately predict popularity across its full range.



The feature importance analysis highlights **acousticness, duration, and danceability** as the most influential predictors in the Random Forest model. These features contribute more to the model's decision-making compared to others, suggesting they contain relatively stronger signals related to popularity.

The importance values are relatively evenly distributed, and no single feature dominates. This shows that popularity is influenced by a combination of factors rather than one defining characteristic.

Some features commonly assumed to be influential, like energy and loudness, show only moderate importance, reinforcing the idea that intuitive assumptions about music attributes do not necessarily translate into predictive power.

5.2 Discussion

The Random Forest model explains approximately **24.5% of the variance** in popularity. While this shows as low, it is consistent with earlier exploratory analysis, which showed weak correlations between individual audio features and the target variable.

The model's performance suggests that audio features contribute to popularity prediction but do not provide a complete explanation. This is further supported by the prediction plot, where the model tends to cluster predictions around mid-range values and struggles to accurately capture extreme popularity scores.

These findings indicate that popularity is influenced by factors beyond audio characteristics alone. External factors like marketing strategies, artist reputation, playlist placement, and listener behavior likely play a significant role in determining a song's success.

6. Conclusion

This project demonstrates that machine learning models can extract meaningful patterns from audio features. These features alone are not sufficient to fully explain song popularity. The Random Forest model improved performance over Linear Regression, but still explained only a limited portion of the variance, confirming that popularity is influenced by factors beyond the available data.

Key Takeaways

- Random Forest outperforms Linear Regression, indicating non-linear relationships
- Audio features have limited predictive power ($R^2 \approx 0.245$)
- Popularity is influenced by external factors beyond audio characteristics

What I Learned

Through this project, I learned how to:

- Build a complete machine learning pipeline
- Analyze and interpret large datasets
- Evaluate model performance beyond numerical metrics
- Understand real-world limitations of predictive modeling

Future Improvements

Future work might include:

- Adding categorical features like genre
- Incorporating artist-level metadata (e.g., popularity, followers)
- Including time-based trends and release information

7. References

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8. Acknowledgement

I used ChatGPT and Google Gemini to assist me with refining report structure, debugging visualization bugs, and reviewing the machine learning pipeline. All final implementation, analysis, and conclusions were completed independently.

9. Source Code

GitHub Repository:
https://github.com/sgyawali990/Spotify-Popularity-Prediction